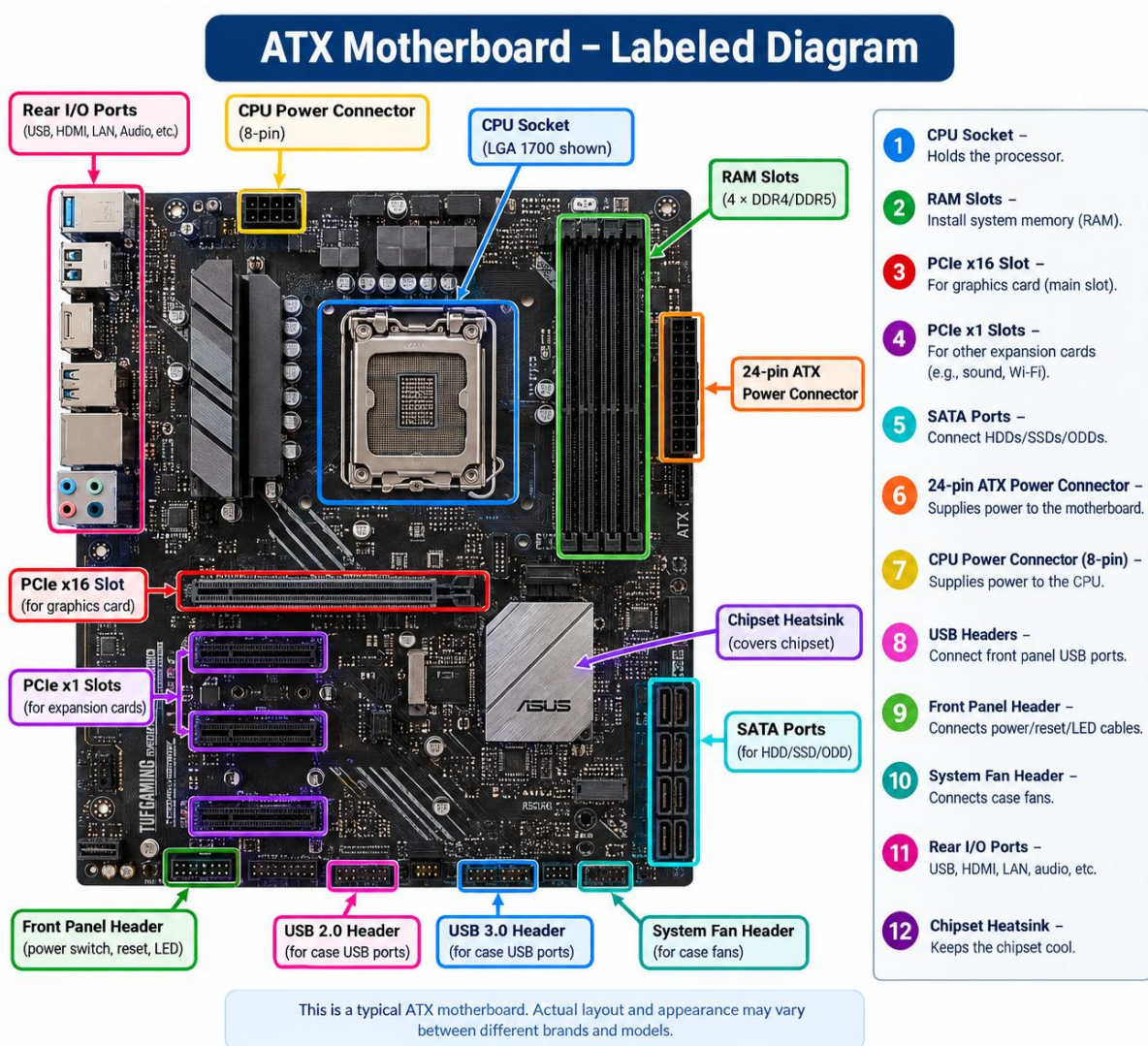


Day 2 : Motherboard and CPU Theory

Task 1 : Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Task 2 : Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

3 Popular CPU Socket Types

A CPU socket is the part of a motherboard where the processor is installed. The CPU and motherboard must have a compatible socket.

1. LGA1151

LGA1151 is an Intel CPU socket used in many older desktop computers. It supports several Intel Core processors.

Example: Intel Core i7-9700K.

2. AM4

AM4 is an AMD CPU socket widely used with Ryzen processors. It supports many popular AMD CPUs.

Example: AMD Ryzen 5 5600.

3. LGA1200

LGA1200 is an Intel socket mainly used for 10th- and 11th-generation desktop processors.

Example: Intel Core i5-10600K.

Socket Type	Company	Example CPU
LGA1151	Intel	Core i7-9700K
AM4	AMD	Ryzen 5 5600
LGA1200	Intel	Core i5-10600K

Task 3 : Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

1. CPU (Central Processing Unit)

The CPU is the brain of the computer. It reads and processes instructions and controls the work of different programs. When you open Spotify, the CPU processes the instructions needed to start the app, load your account, and play music.

2. Chipset

The chipset helps different hardware components communicate with the CPU. It manages connections between the CPU and devices such as the SSD, USB devices, and other motherboard components.

3. How CPU and Chipset Work Together

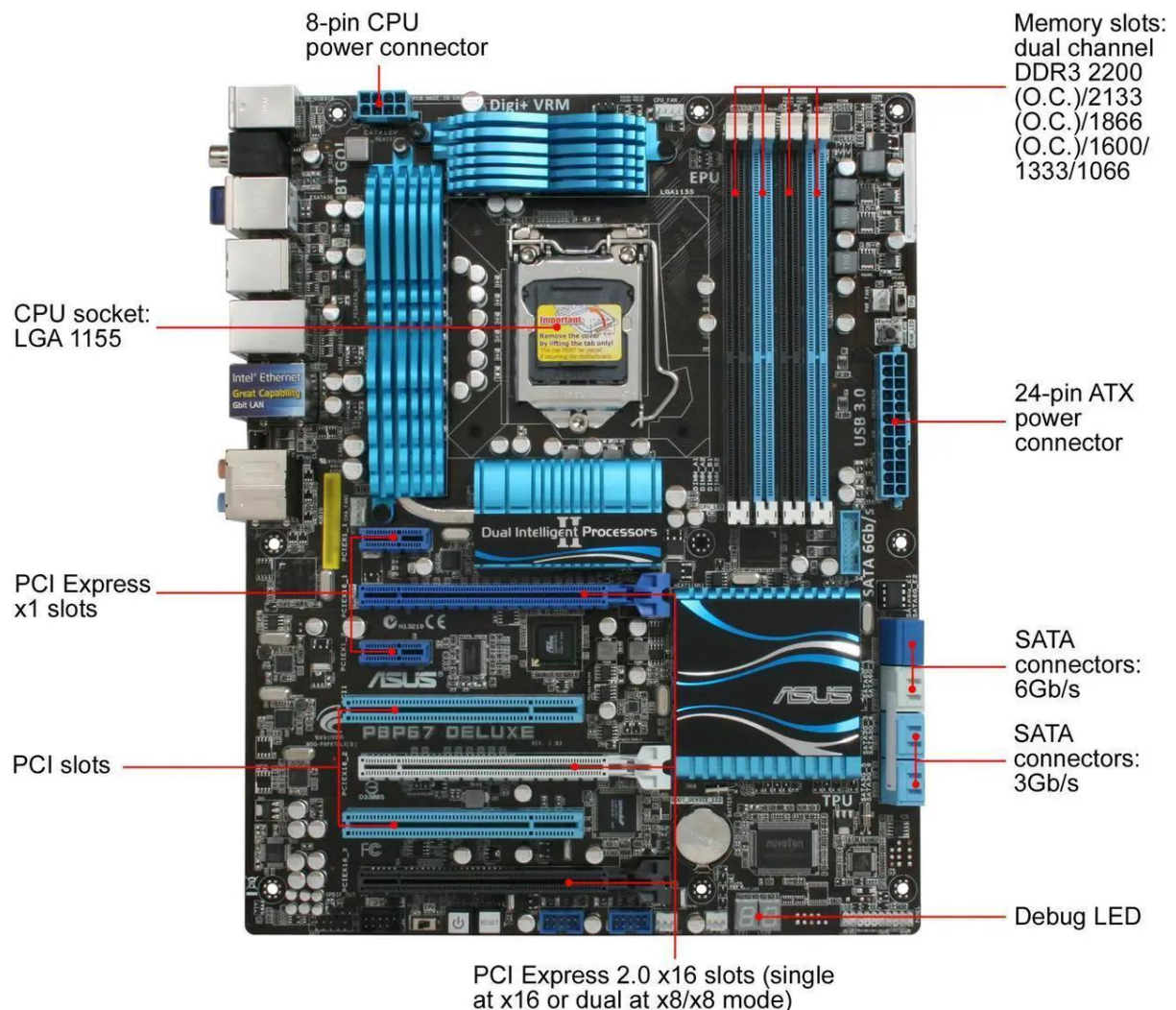
When you open Spotify, the CPU tells the computer what tasks need to be performed. The chipset helps the CPU communicate with the storage drive, RAM, audio hardware, and other devices, allowing Spotify to load and play music smoothly.

Easy to remember:

CPU = Processes instructions

Chipset = Helps hardware communicate

Task 4 : Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



Motherboard Ports and Slots

In the motherboard image, I identified these important ports and slots:

- 1. CPU Socket:** Holds the CPU (processor) on the motherboard.
- 2. RAM Slots:** Used to install RAM modules for temporary data storage.
- 3. PCIe x16 Slot:** Mainly used to install a graphics card (GPU).

4. PCIe x1 Slot: Used for smaller expansion cards, such as sound or Wi-Fi cards.

5. SATA Ports: Used to connect storage devices such as HDDs and SATA SSDs.

6. USB Header: Connects the motherboard to the computer case's front USB ports.

7. 24-Pin ATX Power Connector: Provides power to the motherboard.

8. CPU Power Connector: Provides additional power to the CPU.